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[23CSE104]

FIRST SEMESTER B.Tech EXAMINATION, JANUARY-2025

CYBERSECURITY LANDSCAPE

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		Questions	Blooms Level	CO's	Marks
		PART-A	**5 x2=10**		
1.	**a.**	You are conducting a security audit and find that a company's sensitive information is stored in an unsecured location where it can be accessed by unauthorized individuals. Which element of the **CIA Triad** is most at risk in this situation and explain how is it at risk?	L1	CO1	**02**
	b.	Outline the pros and cons of Biometrics	L2	CO 2	**02**
	c.	Define Vulnerability	L2	CO 3	**02**
	d.	What is a worm	L2	CO 4	**02**
	e.	What is Cyber Risk Assessment	L2	CO 5	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Explain in detail the concept of Parkerian Hexad	L2	CO 1	**05**
		OR			
	b.	Explain the different types of Assets in Cyber Security	L2	CO 1	**05**
3.	**a.**	Illustrate the working of an TLS Handshake	L2	CO 2	**05**
		OR			
	b.	Discuss in detail the hardware tokens types and its advantages	L2	CO 2	**05**
4.	**a.**	Define threats and vulnerabilities in a cyber-safe environment. How are they different?	L2	CO 3	**05**
		OR			
	b.	Illustrate how can organizations minimize the impact of non-adversarial threats, such as natural disasters or human error?	L2	CO 3	**05**
5.	**a.**	Explain the Common Types of DoS Attacks	L2	CO 4	**05**
		OR			
	b.	Outline the lifecycle of an virus	L2	CO 4	**05**

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6.	**a.**	Identify and explain the five types of modern warfare	L2	CO 4	**05**
		OR			
	b.	Outline the working of Token Kidnapping	L2	CO 5	**05**
7.	**a.**	Explain the Risk IT Framework Process Flow in detail	L2	CO 5	**05**
		OR			
	b.	State the advantages of Effective Risk Mitigation	L2	CO 5	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	What are Social Engineering Techniques? Explain at least four types of social engineering attacks and discuss the impact they can have on organizations.	L2	CO 1	**10**
		OR			
	b.	Describe the Crown Jewels in the context of information security?	L2	CO 1	**10**
9.	**a.**	Illustrate the different types of access control models	L2	CO 2	**10**
		OR			
	b.	Explain in detail the identification processes	L2	CO 2	**10**
10.	**a.**	Identify and explain multiple sources of threats, distinguishing between adversarial and non-adversarial threat sources. Provide examples of each.	L2	CO 3	**10**
		OR			
	b.	What are the roles and responsibilities of key individuals in ensuring a cyber-safe environment in an organization?	L2	CO 3	**10**
11.	**a.**	Explain in detail the working of Man in the Middle Attack	L2	CO 4	**10**
		OR			
	b.	Explain in detail how will you consider information an asset	L2	CO 4	**10**
12.	**a.**	Explain in detail the Risk Analysis process	L2	CO 5	**10**
		OR			
	b.	Explain in detail the different types of cost and the benefits of cost analysis in Risk management	L2	CO 5	**10**
13.	**a.**	Explain in detail the Information control (censorship) and it types	L2	CO 5	**10**
		OR			
	b.	Explain Network Authentication, Authorization and Auditing in cyber security with examples	L2	CO 2	**10**